Supplementary files for PhyPro-FM: A Physics and Prompt-Guided Graph Foundation Model for Contingency Analysis

TABLE I
NUMBER OF BENCHMARK TOPOLOGIES AND SAMPLES

Case	Training		Test		No. Samples
	N-1	N-2	N-1	N-2	
39-bus	26	299	7	18	6.0k
118-bus	133	600	35	150	12.9k
300-bus	189	900	51	225	18.7k
1354-bus	300	1200	75	300	25.0k
3012-bus	400	1500	100	375	31.3k

TABLE II
HYPERPARAMETERS OF PROPOSED PHYPRO-FM

Hyperparameter	Value
No. Message Passing Blocks L	6
No. Attention Heads	4
Dropout	0.1
Prompt Length	8
Weight Coefficients $\lambda_V, \lambda_\theta, \lambda_e, \lambda_{d\theta}$	10.0, 1.0, 40.0, 2.0
Optimizer	AdamW
Hidden Units (Backbone)	192
Hidden Units (Correction)	96
Learning Rate (Backbone)	3.0e-4 (Warmup + CosineAnnealing)
Learning Rate (Correction)	5.0e-4
Epoch (Backbone)	100
Epoch (Correction)	50
Batch Size	16
Total Parameters	10.9M

TABLE III
PERFORMANCE OF PHYPRO-FM AND BASELINE METHODS ON NORMAL TOPOLOGY OF DIFFERENT SYSTEMS

A. AVERAGE MEAN ABSOLUTE ERROR OF NODE VOLTAGE MAGNITUDE (P.U.)

Method	39-Bus		118-Bus		300-Bus		1354-Bus		3012-Bus	
	Dedicated	Mixed	Dedicated	Mixed	Dedicated	Mixed	Dedicated	Mixed	Dedicated	Mixed
MLP	6.41e-4	-	1.20e-3	-	6.57e-3	-	5.35e-3	-	4.44e-3	-
GCN	2.83e-3	2.81e-3	7.84e-4	9.32e-4	6.13e-3	6.16e-3	2.65e-3	2.82e-3	2.02e-3	2.31e-3
GAT	9.76e-4	8.24e-4	3.12e-4	4.04e-4	1.14e-3	1.14e-3	7.25e-4	8.04e-4	1.09e-3	1.30e-3
GT	5.11e-4	5.09e-4	1.81e-4	2.32e-4	7.49e-4	8.02e-4	4.48e-4	5.78e-4	8.18e-4	9.87e-4
GridSFM	2.21e-4	2.56e-4	8.34e-5	1.13e-4	3.17e-4	4.30e-4	1.59e-4	2.54e-4	1.84e-4	3.04e-4
GENCO	1.48e-4	9.10e-4	**4.77e-5**	3.87e-4	4.05e-4	1.95e-3	7.15e-4	2.03e-3	3.26e-3	4.79e-3
PhyPro-FM1	1.68e-4	2.09e-4	5.64e-5	1.13e-4	3.22e-4	4.63e-4	1.35e-4	2.12e-4	1.89e-4	3.67e-4
PhyPro-FM2	2.06e-4	1.74e-4	6.86e-5	**8.18e-5**	2.98e-4	**3.07e-4**	1.29e-4	1.60e-4	**1.50e-4**	2.56e-4
PhyPro-FM3	1.52e-4	1.40e-4	9.02e-5	1.17e-4	3.75e-4	4.14e-4	**1.23e-4**	**1.39e-4**	1.72e-4	**2.55e-4**
PhyPro-FM	**1.47e-4**	**1.32e-4**	6.68e-5	8.85e-5	**2.87e-4**	3.45e-4	1.29e-4	1.47e-4	1.89e-4	2.82e-4

B. AVERAGE MEAN ABSOLUTE ERROR OF NODE VOLTAGE PHASE ANGLE (RAD)

Method	39-Bus		118-Bus		300-Bus		1354-Bus		3012-Bus	
	Dedicated	Mixed	Dedicated	Mixed	Dedicated	Mixed	Dedicated	Mixed	Dedicated	Mixed
MLP	6.78e-4	-	1.35e-3	-	3.39e-3	-	1.38e-3	-	1.70e-3	-
GCN	2.70e-3	2.53e-3	2.16e-3	2.74e-3	6.05e-3	6.53e-3	2.07e-3	2.86e-3	3.05e-3	3.90e-3
GAT	1.77e-3	1.89e-3	1.28e-3	1.64e-3	4.77e-3	5.14e-3	1.66e-3	2.33e-3	2.46e-3	2.99e-3
GT	1.53e-3	1.63e-3	1.05e-3	1.40e-3	4.17e-3	4.53e-3	1.39e-3	1.85e-3	2.04e-3	2.47e-3
GridSFM	2.58e-4	3.28e-4	3.83e-4	5.27e-4	9.62e-4	1.74e-3	4.22e-4	6.68e-4	4.68e-4	7.11e-4
GENCO	4.46e-4	3.45e-3	5.84e-4	3.57e-3	2.68e-3	1.15e-2	3.18e-3	7.23e-3	1.31e-2	1.17e-2
PhyPro-FM1	2.03e-4	3.56e-4	2.73e-4	6.38e-4	9.69e-4	1.66e-3	3.94e-4	7.08e-4	4.93e-4	8.25e-4
PhyPro-FM2	2.99e-4	3.03e-4	3.66e-4	4.44e-4	9.73e-4	1.00e-3	3.78e-4	5.19e-4	**4.13e-4**	**6.05e-4**
PhyPro-FM3	**1.91e-4**	**1.84e-4**	**2.40e-4**	**3.29e-4**	**7.36e-4**	**9.40e-4**	**3.34e-4**	**4.33e-4**	4.42e-4	6.17e-4
PhyPro-FM	2.16e-4	2.12e-4	2.70e-4	3.71e-4	8.38e-4	1.08e-3	3.78e-4	4.95e-4	5.06e-4	7.08e-4

C. AVERAGE MEAN ABSOLUTE ERROR OF BRANCH ACTIVE POWER (P.U.)

Method	39-Bus		118-Bus		300-Bus		1354-Bus		3012-Bus	
	Dedicated	Mixed	Dedicated	Mixed	Dedicated	Mixed	Dedicated	Mixed	Dedicated	Mixed
MLP	2.92e-2	-	6.64e-3	-	3.76e-2	-	8.45e-2	-	4.01e-2	-
GCN	9.07e-2	8.82e-2	1.68e-2	2.05e-2	7.92e-2	9.05e-2	0.263	0.368	0.415	0.675
GAT	5.71e-2	6.33e-2	1.02e-2	1.36e-2	5.62e-2	6.89e-2	0.254	0.367	0.392	0.545
GT	5.38e-2	5.86e-2	9.69e-3	1.28e-2	5.79e-2	7.03e-2	0.224	0.295	0.316	0.415
GridSFM	1.73e-2	2.01e-2	5.20e-3	5.53e-3	2.14e-2	2.81e-2	7.30e-2	0.109	0.166	0.229
GENCO	2.53e-2	0.14	5.18e-3	4.30e-2	5.54e-2	0.268	1.03	1.96	0.85	1.53
PhyPro-FM1	**2.00e-3**	3.27e-3	**6.07e-4**	1.24e-3	1.85e-3	2.76e-3	**3.06e-3**	**5.01e-3**	**2.35e-3**	4.16e-3
PhyPro-FM2	9.28e-3	8.90e-3	2.12e-3	2.46e-3	5.93e-3	6.49e-3	1.72e-2	2.51e-2	1.57e-2	2.44e-2
PhyPro-FM3	3.38e-3	3.26e-3	9.19e-4	1.24e-3	2.73e-3	3.29e-3	5.86e-3	6.65e-3	3.91e-3	5.54e-3
PhyPro-FM	2.57e-3	**2.55e-3**	7.11e-4	**1.00e-3**	**1.82e-3**	**2.29e-3**	4.23e-3	5.24e-3	2.90e-3	**3.99e-3**

D. AVERAGE MAXIMUM ABSOLUTE ERROR OF BRANCH ACTIVE POWER (P.U.)

Method	39-Bus		118-Bus		300-Bus		1354-Bus		3012-Bus	
	Dedicated	Mixed	Dedicated	Mixed	Dedicated	Mixed	Dedicated	Mixed	Dedicated	Mixed
MLP	0.138	-	8.17e-2	-	3.66	-	8.73	-	11.9	-
GCN	0.422	0.478	0.32	0.385	4.91	4.69	18.9	24.1	58.4	90
GAT	0.292	0.349	0.249	0.294	2.58	3.27	20	27.4	50.3	64.7
GT	0.355	0.325	0.268	0.308	2.47	3.98	20	23	30.5	42.3
GridSFM	9.86e-2	0.122	0.118	0.107	1.32	2.03	2.86	4.07	20.1	25.9
GENCO	0.168	0.798	9.25e-2	0.626	2.78	19.6	61.7	93.7	41.5	97.9
PhyPro-FM1	**8.96e-3**	1.43e-2	**7.01e-3**	1.39e-2	2.36e-2	3.69e-2	**4.52e-2**	**8.22e-2**	**5.70e-2**	8.82e-2
PhyPro-FM2	3.71e-2	3.70e-2	2.79e-2	2.94e-2	8.59e-2	9.94e-2	0.389	0.664	0.523	0.771
PhyPro-FM3	1.49e-2	1.48e-2	1.07e-2	1.40e-2	9.48e-2	0.11	0.158	0.149	0.24	0.356
PhyPro-FM	1.08e-2	**1.13e-2**	7.96e-3	**1.03e-2**	**2.19e-2**	**2.76e-2**	6.65e-2	8.83e-2	6.50e-2	**8.05e-2**

TABLE IV

PERFORMANCE OF PHYPRO-FM AND BASELINE METHODS ON SEEN N-1/N-2 OF DIFFERENT SYSTEMS (DEDICATED TRAINING)

A. AVERAGE MEAN ABSOLUTE ERROR OF NODE VOLTAGE MAGNITUDE (P.U.)

Method	39-Bus		118-Bus		300-Bus		1354-Bus		3012-Bus	
	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2
MLP	7.14e-4	9.34e-4	1.51e-3	1.81e-3	6.68e-3	6.78e-3	5.40e-3	5.42e-3	4.50e-3	4.54e-3
GCN	3.39e-3	3.64e-3	8.35e-4	9.21e-4	6.13e-3	6.23e-3	2.64e-3	2.66e-3	2.04e-3	2.08e-3
GAT	1.11e-3	1.24e-3	3.46e-4	3.72e-4	1.16e-3	1.21e-3	7.21e-4	7.41e-4	1.10e-3	1.13e-3
GT	5.99e-4	7.20e-4	2.03e-4	2.22e-4	7.61e-4	8.04e-4	4.37e-4	4.56e-4	8.28e-4	8.51e-4
GridSFM	2.74e-4	3.37e-4	1.06e-4	1.27e-4	3.19e-4	3.37e-4	1.67e-4	1.77e-4	1.95e-4	2.10e-4
GENCO	1.99e-4	2.36e-4	**5.79e-5**	**6.77e-5**	4.19e-4	4.46e-4	7.22e-4	7.44e-4	3.27e-3	3.33e-3
PhyPro-FM1	1.64e-4	2.02e-4	6.40e-5	7.33e-5	3.05e-4	3.17e-4	1.40e-4	1.43e-4	1.95e-4	2.04e-4
PhyPro-FM2	2.23e-4	2.67e-4	6.96e-5	7.75e-5	2.66e-4	2.74e-4	1.16e-4	1.17e-4	**1.46e-4**	**1.50e-4**
PhyPro-FM3	1.76e-4	2.13e-4	1.00e-4	1.11e-4	3.59e-4	3.67e-4	1.25e-4	1.27e-4	1.73e-4	1.78e-4
PhyPro-FM	**1.47e-4**	**1.83e-4**	6.48e-5	7.31e-5	**2.58e-4**	**2.66e-4**	**1.15e-4**	**1.16e-4**	1.71e-4	1.75e-4

B. AVERAGE MEAN ABSOLUTE ERROR OF NODE VOLTAGE PHASE ANGLE (RAD)

Method	39-Bus		118-Bus		300-Bus		1354-Bus		3012-Bus	
	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2
MLP	8.21e-4	1.04e-3	1.72e-3	2.03e-3	4.04e-3	4.24e-3	1.40e-3	1.57e-3	1.84e-3	1.82e-3
GCN	3.05e-3	4.10e-3	2.73e-3	3.12e-3	7.38e-3	7.73e-3	2.13e-3	2.25e-3	3.11e-3	3.20e-3
GAT	2.16e-3	3.02e-3	1.66e-3	1.93e-3	5.91e-3	6.12e-3	1.71e-3	1.85e-3	2.56e-3	2.61e-3
GT	1.84e-3	2.45e-3	1.35e-3	1.60e-3	5.05e-3	5.06e-3	1.42e-3	1.56e-3	2.16e-3	2.19e-3
GridSFM	3.63e-4	5.31e-4	4.83e-4	5.61e-4	8.74e-4	9.08e-4	4.13e-4	4.58e-4	4.66e-4	4.90e-4
GENCO	6.41e-4	8.84e-4	7.24e-4	8.31e-4	3.16e-3	3.03e-3	3.27e-3	3.39e-3	1.33e-2	1.31e-2
PhyPro-FM1	2.99e-4	3.98e-4	2.99e-4	3.57e-4	9.89e-4	9.18e-4	4.15e-4	4.94e-4	5.08e-4	5.33e-4
PhyPro-FM2	3.77e-4	4.80e-4	3.56e-4	4.14e-4	7.45e-4	7.47e-4	**3.24e-4**	**3.36e-4**	**3.58e-4**	**3.64e-4**
PhyPro-FM3	2.62e-4	3.45e-4	2.50e-4	2.97e-4	6.49e-4	6.63e-4	3.26e-4	3.48e-4	4.33e-4	4.46e-4
PhyPro-FM	**2.59e-4**	**3.43e-4**	**2.49e-4**	**2.94e-4**	**6.47e-4**	**6.60e-4**	3.25e-4	3.47e-4	4.40e-4	4.51e-4

C. Average Mean Absolute Error of Branch Active Power (p.u.)

Method	39-Bus		118-Bus		300-Bus		1354-Bus		3012-Bus	
	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2
MLP	3.55e-2	4.40e-2	8.05e-3	9.43e-3	3.97e-2	4.08e-2	8.53e-2	8.62e-2	4.09e-2	4.14e-2
GCN	0.114	0.142	2.02e-2	2.33e-2	8.64e-2	8.90e-2	0.258	0.266	0.41	0.422
GAT	7.49e-2	9.93e-2	1.27e-2	1.52e-2	6.41e-2	6.78e-2	0.251	0.261	0.39	0.399
GT	7.09e-2	8.92e-2	1.18e-2	1.39e-2	6.95e-2	7.67e-2	0.221	0.231	0.315	0.325
GridSFM	2.50e-2	3.49e-2	6.80e-3	8.14e-3	2.55e-2	2.80e-2	7.85e-2	8.25e-2	0.173	0.183
GENCO	3.29e-2	3.92e-2	6.14e-3	7.06e-3	6.14e-2	6.43e-2	1.04	1.05	0.85	0.85
PhyPro-FM1	**2.78e-3**	**3.78e-3**	7.12e-4	8.38e-4	2.01e-3	2.12e-3	**3.21e-3**	**3.41e-3**	**2.41e-3**	**2.50e-3**
PhyPro-FM2	1.10e-2	1.42e-2	2.30e-3	2.55e-3	6.14e-3	6.25e-3	1.77e-2	1.77e-2	1.50e-2	1.52e-2
PhyPro-FM3	4.27e-3	5.36e-3	1.02e-3	1.15e-3	2.81e-3	2.90e-3	5.91e-3	5.98e-3	3.94e-3	4.00e-3
PhyPro-FM	3.03e-3	3.88e-3	**6.89e-4**	**8.00e-4**	**1.67e-3**	**1.74e-3**	3.71e-3	3.75e-3	2.59e-3	2.63e-3

D. Average Maximum Absolute Error of Branch Active Power (p.u.)

Method	39-Bus		118-Bus		300-Bus		1354-Bus		3012-Bus	
	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2
MLP	0.19	0.241	0.102	0.124	3.73	3.79	8.99	8.82	12.4	12.7
GCN	0.555	0.719	0.362	0.425	5.19	5.32	15.8	17.7	57.8	60.8
GAT	0.4	0.554	0.3	0.378	2.85	2.92	17.1	19.3	51.3	52.6
GT	0.407	0.516	0.302	0.365	3.42	3.85	16.9	18	31.5	33.3
GridSFM	0.144	0.224	0.149	0.176	1.65	1.76	3.33	3.57	21.3	23
GENCO	0.213	0.238	0.107	0.129	3.17	3.22	58.4	60.9	41.3	41.3
PhyPro-FM1	**1.33e-2**	1.69e-2	7.86e-3	9.25e-3	2.57e-2	2.63e-2	**5.21e-2**	6.08e-2	5.89e-2	6.27e-2
PhyPro-FM2	4.62e-2	6.01e-2	2.80e-2	3.12e-2	8.34e-2	8.37e-2	0.422	0.426	0.471	0.484
PhyPro-FM3	1.85e-2	2.40e-2	1.16e-2	1.24e-2	9.05e-2	9.42e-2	0.156	0.16	0.256	0.25
PhyPro-FM	1.40e-2	**1.65e-2**	**7.08e-3**	**8.45e-3**	**1.96e-2**	**2.14e-2**	5.83e-2	**6.03e-2**	**5.61e-2**	**5.73e-2**

TABLE V

Performance of PhyPro-FM and Baseline Methods on Seen N-1/N-2 of Different Systems (Mixed Training)

A. Average Mean Absolute Error of Node Voltage Magnitude (p.u.)

Method	39-Bus		118-Bus		300-Bus		1354-Bus		3012-Bus	
	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2
GCN	3.26e-3	3.48e-3	9.97e-4	1.11e-3	6.18e-3	6.28e-3	2.82e-3	2.84e-3	2.33e-3	2.37e-3
GAT	9.71e-4	1.16e-3	4.58e-4	5.06e-4	1.17e-3	1.22e-3	7.99e-4	8.26e-4	1.32e-3	1.35e-3
GT	6.17e-4	7.35e-4	2.70e-4	2.97e-4	8.15e-4	8.65e-4	5.74e-4	5.98e-4	1.00e-3	1.03e-3
GridSFM	2.76e-4	3.04e-4	1.37e-4	1.61e-4	4.13e-4	4.35e-4	2.66e-4	2.79e-4	3.22e-4	3.46e-4
GENCO	1.12e-3	1.43e-3	4.31e-4	4.82e-4	2.00e-3	2.06e-3	2.01e-3	2.06e-3	4.80e-3	4.87e-3
PhyPro-FM1	2.25e-4	2.85e-4	1.34e-4	1.49e-4	4.60e-4	4.78e-4	2.17e-4	2.22e-4	3.75e-4	3.88e-4
PhyPro-FM2	1.74e-4	2.07e-4	**8.56e-5**	**9.25e-5**	**2.84e-4**	**2.87e-4**	1.47e-4	1.48e-4	**2.36e-4**	**2.39e-4**
PhyPro-FM3	1.50e-4	1.86e-4	1.27e-4	1.39e-4	4.02e-4	4.11e-4	1.42e-4	1.42e-4	2.57e-4	2.61e-4
PhyPro-FM	**1.26e-4**	**1.59e-4**	8.69e-5	9.69e-5	3.14e-4	3.25e-4	**1.32e-4**	**1.32e-4**	2.55e-4	2.58e-4

B. Average Mean Absolute Error of Node Voltage Phase Angle (rad)

Method	39-Bus		118-Bus		300-Bus		1354-Bus		3012-Bus	
	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2
GCN	2.92e-3	4.01e-3	3.45e-3	4.07e-3	7.96e-3	8.40e-3	2.91e-3	3.04e-3	3.91e-3	4.00e-3
GAT	2.40e-3	3.18e-3	2.17e-3	2.51e-3	6.38e-3	6.65e-3	2.38e-3	2.53e-3	3.04e-3	3.13e-3
GT	1.95e-3	2.56e-3	1.81e-3	2.16e-3	5.43e-3	5.56e-3	1.88e-3	2.03e-3	2.55e-3	2.62e-3
GridSFM	4.39e-4	5.51e-4	6.00e-4	6.86e-4	1.16e-3	1.24e-3	6.54e-4	7.31e-4	6.98e-4	7.29e-4
GENCO	3.84e-3	5.25e-3	4.55e-3	5.23e-3	1.29e-2	1.34e-2	7.39e-3	7.50e-3	1.19e-2	1.17e-2
PhyPro-FM1	4.48e-4	5.85e-4	7.47e-4	8.30e-4	1.84e-3	1.71e-3	7.09e-4	8.39e-4	8.63e-4	8.89e-4
PhyPro-FM2	2.96e-4	3.93e-4	4.47e-4	5.13e-4	**8.11e-4**	**8.00e-4**	4.36e-4	4.66e-4	**5.30e-4**	**5.37e-4**
PhyPro-FM3	2.60e-4	2.94e-4	3.88e-4	**4.33e-4**	8.94e-4	8.62e-4	**4.19e-4**	**4.38e-4**	6.16e-4	6.19e-4
PhyPro-FM	**2.50e-4**	**2.88e-4**	**3.83e-4**	4.36e-4	8.99e-4	8.64e-4	4.20e-4	4.41e-4	6.27e-4	6.27e-4

C. AVERAGE MEAN ABSOLUTE ERROR OF BRANCH ACTIVE POWER (P.U.)

Method	39-Bus		118-Bus		300-Bus		1354-Bus		3012-Bus	
	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2
GCN	0.115	0.138	2.50e-2	2.80e-2	9.60e-2	9.82e-2	0.365	0.372	0.671	0.681
GAT	8.68e-2	0.106	1.66e-2	1.96e-2	7.83e-2	8.26e-2	0.364	0.373	0.541	0.553
GT	7.65e-2	9.59e-2	1.52e-2	1.89e-2	8.26e-2	9.13e-2	0.291	0.303	0.411	0.424
GridSFM	2.93e-2	3.83e-2	7.21e-3	8.64e-3	3.27e-2	3.59e-2	0.115	0.12	0.236	0.248
GENCO	0.152	0.174	4.54e-2	4.86e-2	0.278	0.28	1.96	1.97	1.53	1.53
PhyPro-FM1	4.26e-3	5.71e-3	1.50e-3	1.69e-3	3.14e-3	3.32e-3	5.18e-3	5.40e-3	4.23e-3	4.34e-3
PhyPro-FM2	9.74e-3	1.18e-2	2.63e-3	2.85e-3	6.53e-3	6.61e-3	2.60e-2	2.61e-2	2.34e-2	2.35e-2
PhyPro-FM3	4.09e-3	4.86e-3	1.37e-3	1.50e-3	3.39e-3	3.47e-3	6.73e-3	6.78e-3	5.59e-3	5.65e-3
PhyPro-FM	**2.97e-3**	**3.60e-3**	**9.78e-4**	**1.10e-3**	**2.09e-3**	**2.17e-3**	**4.64e-3**	**4.68e-3**	**3.58e-3**	**3.61e-3**

D. AVERAGE MAXIMUM ABSOLUTE ERROR OF BRANCH ACTIVE POWER (P.U.)

Method	39-Bus		118-Bus		300-Bus		1354-Bus		3012-Bus	
	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2	N-1	N-2
GCN	0.614	0.741	0.469	0.52	4.85	4.99	21.8	22.6	88.3	91.2
GAT	0.582	0.701	0.336	0.429	3.95	4.16	24.9	26.2	66	67.6
GT	0.42	0.52	0.338	0.468	5.33	6.47	20.4	22	43.1	45
GridSFM	0.186	0.241	0.132	0.154	2.37	2.49	4.57	5.18	27.1	29.6
GENCO	0.8	0.984	0.67	0.698	20.5	20.7	93.6	94.6	98.7	98.7
PhyPro-FM1	1.96e-2	2.50e-2	1.66e-2	1.82e-2	4.32e-2	4.34e-2	9.01e-2	9.96e-2	9.05e-2	9.64e-2
PhyPro-FM2	4.10e-2	5.06e-2	3.03e-2	3.33e-2	9.31e-2	9.17e-2	0.713	0.711	0.688	0.7
PhyPro-FM3	1.75e-2	2.17e-2	1.56e-2	1.60e-2	0.107	0.106	0.154	0.153	0.364	0.367
PhyPro-FM	**1.30e-2**	**1.64e-2**	**9.94e-3**	**1.09e-2**	**2.50e-2**	**2.66e-2**	**7.69e-2**	**7.97e-2**	**7.22e-2**	**7.30e-2**

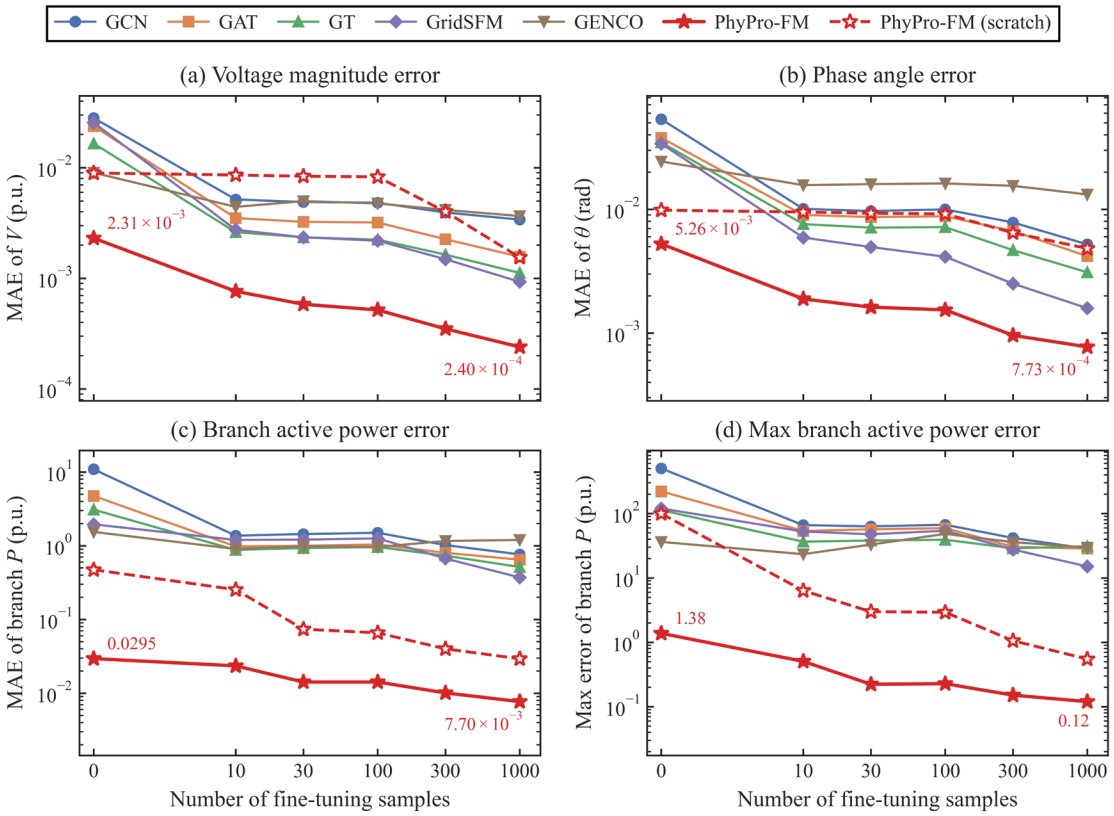


Fig. 1. Fine-tuning performance of different methods on normal topology in 1354-bus system.

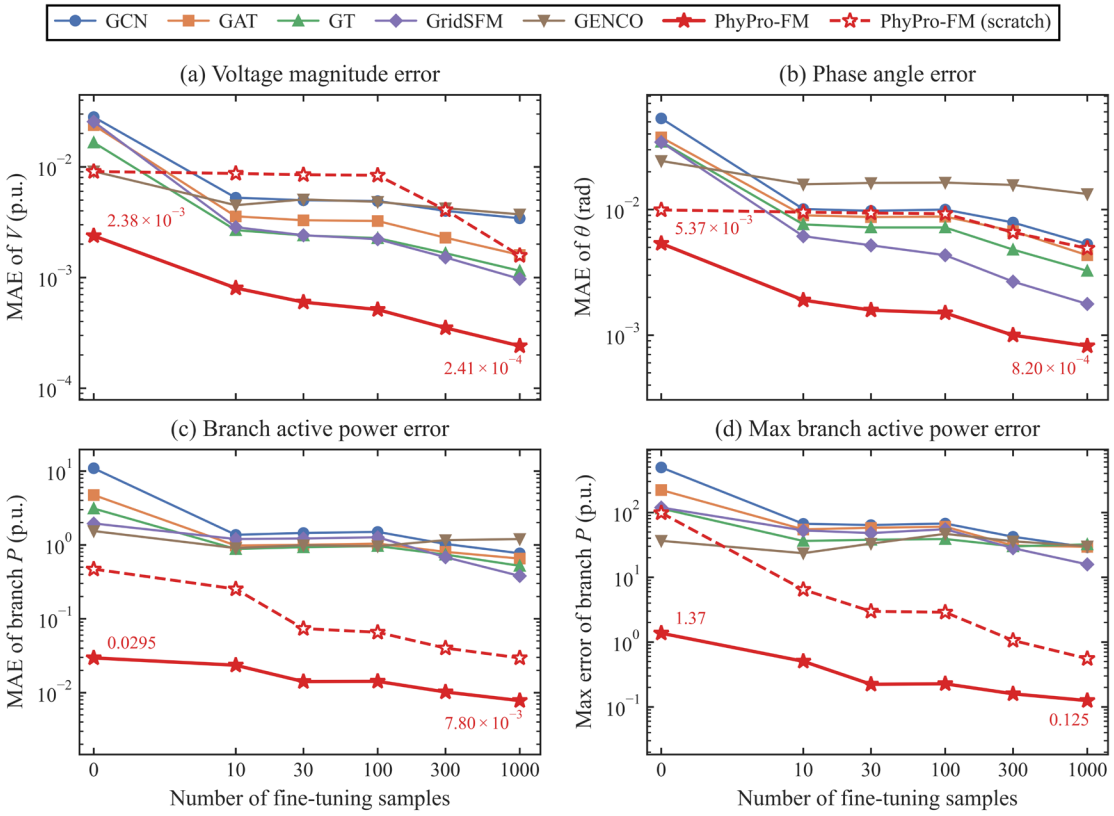


Fig. 2. Fine-tuning performance of different methods on seen N-1 scenarios in 1354-bus system.

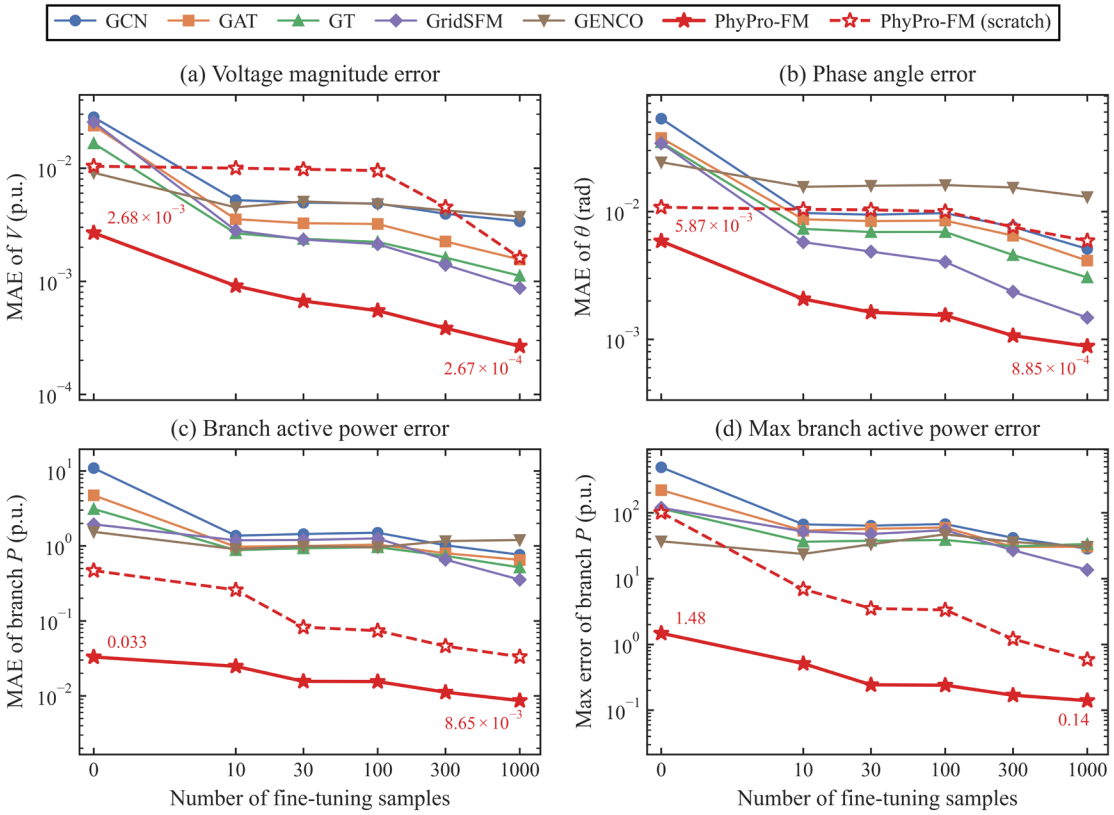


Fig. 3. Fine-tuning performance of different methods on seen N-2 scenarios in 1354-bus system.